

Theory of Computation

BSc Computer Science - Durham University

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1 Module Introduction

Theory of Computation has the module code COMP2181 and is led by Dr Barnaby Martin.

1.1 Assessment

This module is assessed through two summative benchtests, and one final exams, these can be seen in Table 1. **Final dates to be confirmed.**

Title	Opening	Due	Feedback	Professor	Weight
Benchtest 1	12/01/2027	12/01/2027	09/02/2027	Dr Dantchev and Dr Mertziros	17%
Benchtest 2	19/03/2027	19/03/2027	07/05/2027	Dr Dantchev and Dr Mertziros	17%
Final Exam	May Exam	n/a	n/a	Dr Barnaby Martin	66%

Table 1: Assessments for Theory of Computation.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2. **Final times and locations to be confirmed.**

Day	Time	Class Type	Location	Lecturer
Tuesday	10:00 - 11:00	Lecture	CLC202	Dr Barnaby Martin
Wednesday	09:00 - 11:00	Problem Class	TLC033	n/a
Thursday	14:00 - 15:00	Lecture	CLC202	Dr Sukanya Pandey

Table 2: Michaelmas Term Timetable for Theory of Computation.

2 Models of Computation

3 Algorithms and Complexity

Part I

Appendix

4 Bibliography

4.1 Module Professors

Module Lead: Doctor Barnaby Martin

- Focus: Complexity theory, finite model theory, and the links between logic and complexity.
- Email: barnaby.d.martin@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Sukanya Pandey

- Focus: Not specified.
- Email: sukanya.pandey@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor Stefan Dantchev

- Focus: Combinatorics.
- Email: s.s.dantchev@durham.ac.uk
- Office Hours: **To be confirmed.**

Lecturer: Doctor George Mertzios

- Focus: Algorithms and networks.
- Email: george.mertzios@durham.ac.uk
- Office Hours: **To be confirmed.**

4.2 References

4.3 List of Figures